

PJP-14: A Universal Lossless Compression Framework with 256 Elementary Transforms, 65 535 Transform Pairs, and Deep Multi-Pair Search

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Abstract

We present a universal lossless compression framework that reformulates compression as an optimisation problem over a large, structured set Θ of bijective byte-level preprocessing transforms. For an input byte sequence $\mathbf{x} \in \mathbb{B}^n$ ($\mathbb{B} = \{0, \dots, 255\}$), the compressor evaluates every transform $\tau \in \Theta$, compresses $\tau(\mathbf{x})$ with a backend codec C , and selects the pipeline that minimises $|C(\tau(\mathbf{x}))|$ while ensuring perfect reversibility. The set Θ contains 65 792 paths: the raw identity, 256 single elementary bijections, and 65 535 ordered pairs of transforms drawn from the full set of 256 elementary transforms (excluding the identity–identity pair). Advanced modes extend this to sequences of up to three pairs, yielding a search space exceeding 281 trillion variations. Each elementary transform is explicitly invertible; their composition is automatically invertible. The chosen path is encoded in a variable-length header of 1–3 bytes (up to $2 + 2k$ bytes for multi-pair). The framework integrates hybrid dictionary tokenisation, quantum-inspired permutations, a LZ77+Huffman pipeline, and multiple backend compressors (Zstandard, PAQ, raw) with automatic fallback. Exhaustive self-tests verify the bijectivity of every transform and every composed pair on all 256 byte values, guaranteeing losslessness by construction.

1. Introduction

Traditional compression systems apply a fixed entropy coder directly to the input sequence. PJP-14 introduces a reversible preprocessing stage τ that transforms $\mathbf{X}$ into $\mathbf{Y} = \tau(\mathbf{X})$ before entropy coding. The transform is chosen from a large, structured family of bijections:

$$\tau^* = \arg\min_{\tau \in \Theta} |C(\tau(\mathbf{X}))|,$$

subject to the round-trip condition $C^{-1}(C(\tau(\mathbf{X}))) = \tau(\mathbf{X})$ and $\tau^{-1}(\tau(\mathbf{X})) = \mathbf{X}$. By searching over Θ and picking the shortest compressed output, the system adapts to the statistical structure of the input without losing any information.

Our contribution is a unified implementation that combines:

- 256 invertible byte-level transforms (numbered 1–256), including XOR masks, bit rotations, arithmetic operations, finite-field substitutions, Huffman coding, a run-length/block-run encoder, a prefix-free nibble coder (Constant Diapason), dictionary tokenisers, and two new transforms: 4-byte pattern XOR (#57) and Calculus2⁶⁴ with Gamma coding (#58).
- 65 535 ordered pairs of these transforms (all combinations except the identity pair), providing a rich set of two-stage transformations.
- Advanced “Deep Ultra” and “Ultra++” modes that explore sequences of up to three transform pairs, exploiting a search space of $65,535^3 \approx 2.81 \times 10^{14}$ possibilities.
- A compact header encoding that distinguishes single transforms, transform pairs, and multi-pair sequences.
- Multiple compression backends (Zstandard, PAQ, raw) with automatic fallback and verification; an optional LZ77+Huffman pipeline can replace the backend codec.
- Hybrid dictionary compression for textual data (static, dynamic, line-based).
- Optional quantum-inspired permutations generated from Qiskit circuits, which can be added as extra transforms (indices 257+).

- Built-in exhaustive self-tests that confirm every transform and every composed path is bijective on all 256 byte values.

The framework is designed to be provably lossless, with rigorous verification ensuring bit-perfect reconstruction for all inputs.

2. Mathematical Foundations

2.1 Basic Notation

Let $\mathbb{B} = \{0, 1, \dots, 255\}$ be the set of bytes. A file is a vector $\mathbf{X} = (x_0, x_1, \dots, x_{n-1}) \in \mathbb{B}^n$. Elementary operations include:

- Bitwise XOR: $\oplus$
- Left/right shifts: $\ll, \gg$
- Cyclic rotation: $\text{rotl}_k, \text{rotr}_k$
- Addition/subtraction modulo 256: $\pm_{256}$

A transform is a function $T : \mathbb{B}^n \rightarrow \mathbb{B}^n$ for which there exists an inverse T^{-1} such that $\forall \mathbf{X} : T^{-1}(T(\mathbf{X})) = \mathbf{X}$. Composition $(T_a \circ T_b)(\mathbf{X}) = T_a(T_b(\mathbf{X}))$, inverse $T_b^{-1} \circ T_a^{-1}$.

2.2 Families of Elementary Transforms (Exact Order in PJP-14)

The framework defines 256 base transforms (1–256). We list them grouped by category, exactly as they appear in the implementation.

1. Prime-XOR transforms (1–3)

- Transform 1: XORs every third byte with a key derived from all primes < 256, repeated repeat_count times. Self-inverse.
- Transform 2: Computes a pattern index from a checksum, XORs every fourth byte (starting at index 1) with a 4-element pattern; stores the index in a 1-byte header.
- Transform 3: Bit-rotates every fifth byte (starting at index 2) by a rotation derived from data length and sum; stores rotation in a 1-byte header.

2. Index-dependent arithmetic (4–6)

- Transform 4: Subtracts $i \bmod 256$ from byte i , repeated repeat_count times.
- Transform 5: Rotates all bytes left by 3 bits.
- Transform 6: Applies a fixed substitution table (Fisher–Yates shuffle seeded with 42).

3. PI-digit and nearest-prime XOR (7–9)

- Transform 7: XORs all bytes with a byte derived from the data length, then cyclically with a rotation of the hardcoded PI-digits array; self-inverse.
- Transform 8: Similar to 7 but replaces the first XOR with the nearest prime around $|\mathbf{x}| \bmod 256$; self-inverse.
- Transform 9: Combines the above with a seed-table value; self-inverse.

4. Pattern-counting XOR (10)

- Transform 10: Counts occurrences of the substring “X1”, derives an XOR byte from that count, stores it in a 1-byte header.

5. Fibonacci-XOR (11–12)

- Transform 11: XORs each byte with a key that mixes a Fibonacci number and a position-dependent value; self-inverse.
- Transform 12: XORs each byte with the Fibonacci sequence modulo 256; self-inverse.

6. Prime-repeat XOR (13)

- Transform 13: Finds the nearest prime after $|\mathbf{x}| \bmod 256$ iteratively repeats times, then XORs all bytes with the final prime; stores repeats-1 in a header byte.

7. Checksum append (14)

- Transform 14: Appends the sum of all bytes modulo 256; inverse drops the last byte.

8. Pattern addition (15)

- Transform 15: Adds a 3-element pattern (mod 256) to every third byte starting at index 0; stores pattern index in header.

9. Constant-derived masks (16–21)

- Transform 16: XORs every byte with $(|\mathbf{X}| \cdot 7 + 13) \bmod 256$; self-inverse.
- Transform 17: Uses a π -derived k to build a mask; self-inverse.
- Transform 18: Mask from Basel problem digits ($\pi^2/6$); self-inverse.
- Transform 19: Mask from digits of $1/e$; self-inverse.
- Transform 20: Mask from digits of $5e$; self-inverse.
- Transform 21: Adds 255 modulo 256 to every byte; inverse subtracts 255.

10. Dictionary and text tokenisation (22–27)

- Transform 22: Base64 encoding (self-inverse via decode).
- Transform 23: SHA-256 word tokeniser: replaces repeated words with 8-byte hashes; stores a dictionary.
- Transform 24: XOR-prime-hash word tokeniser, analogous to #23.
- Transform 25: Dynamic dictionary tokeniser: splits text into syntactic chunks, builds a frequency dictionary, replaces chunks with 2-byte indices.
- Transform 26: SHA-256 block masking: for each 1024-byte block, XORs with the SHA-256 of a secret and block number; self-inverse.
- Transform 27: 6-bit text packing: if all characters belong to a 64-char alphabet, packs them into 6-bit codes with a 4-byte length prefix; otherwise stores raw. Fully reversible.

11. 3-Byte subtraction (28–30)

- Transform 28: Groups data into 3-byte integers, subtracts a per-block key $K_j = (j \cdot 65537 + 12345) \bmod 2^{16}$ modulo 2^{24} ; stores pad length in header.
- Transform 29: Searches for a global 16-bit key that minimises variance of the transformed blocks; stores the chosen key in a 2-byte header.

- Transform 30: Uses a heuristic to find a 24-bit key; stores the 3-byte key in header.

12. Identity placeholders (31–32)

- Transforms 31–32: Currently identity (reserved for DOCX-aware transforms).

13. Constant Diapason (33)

- Transform 33: Prefix-free nibble coding; applies iteratively until the bit-length stops shrinking. Inverse decodes the codewords.

14. Block-Run Encoding (34)

- Transform 34: Encodes runs of identical bytes (up to length 43) using a 1-bit flag, byte value, and 6-bit run length; non-run chunks are stored raw with length. Inverse reconstructs the original stream.

15. Finite-field transforms (35–40)

- Transforms 35–40: Apply exponentiation in $\mathbb{F}_{257}$. Transform 35 uses a fixed exponent $n=3$; 36 uses a key derived from data length; 37 operates on 1024-byte blocks with per-block keys; 38–40 add backend compression of the transformed blocks. All are invertible via modular inverse of the exponent.

16. Special fixed-pattern XOR and Huffman (41–47)

- Transform 41: XORs the first 8 bytes with the mask [0x27, 0x03]; self-inverse.
- Transform 42: XORs all bytes with the same 2-byte mask cyclically; self-inverse.
- Transform 43: XORs every third byte with the 3-byte mask [0x10, 0x00, 0x00]; self-inverse.
- Transform 44: Base64 encoding (identical to #22, provided as a convenience).
- Transform 45: Canonical Huffman coding: builds a Huffman tree from byte frequencies and encodes the data; stores tree lengths in a 256-byte header. Inverse decodes the bitstream.
- Transform 46: XORs each byte with a pre-computed 100-byte mask (power-of-2 based); self-inverse.
- Transform 47: XORs each byte with the first element of a modified PAQ state table row; self-inverse.

17. Dynamic seed XOR (48–56, 59–255)

- Transforms 48–56 and 59–255: For transform index t , a seed is retrieved from a 126-row table using t and the data length. All bytes are XORed with this seed. Self-inverse.

18. New specialised transforms (57–58)

- Transform 57 (4-byte pattern XOR): Finds the most frequent 4-byte integer (little-endian) in the data, XORs every 4-byte block with that key, and stores the key (4 bytes) plus a padding-length byte in the header. The most common pattern becomes zeros. Inverse XORs with the stored key and removes padding. Self-inverse (since XOR is).

- Transform 58 (Calculus2⁶⁴): Pads data to a multiple of 8 bytes, interprets it as 64-bit integers, searches for a 32-bit key such that subtracting it from each integer (mod 2⁶⁴) yields small values that can be compactly represented with Gamma codes. The key is found via a quantum-inspired search over 2³³ seeds within a time limit. The header stores the original length (8 bytes), the key (4 bytes), and a pad-length byte. The inverse decodes the Gamma stream, adds the key, and trims to the original length.

19. Identity (256)

- Transform 256: $T_{256}(\mathbf{X}) = \mathbf{X}$.

All transforms are implemented with explicit inverse functions. The code includes a self-test that verifies $T^{-1}(T([b]))=[b]$ for every byte b and every transform T in the set.

3. Transform Pairs and the Search Space

Composing two elementary transforms yields a more powerful bijection. The set of candidate paths is:

$$\Theta = \{\text{raw}\} \cup \{T_t \mid 1 \leq t \leq 256\} \cup \{T_a \circ T_b \mid 1 \leq a, b \leq 256, (a, b) \neq (256, 256)\}.$$

Thus $|\Theta| = 1 + 256 + 65,535 = 65,792$. Each ordered pair (a, b) is assigned a linear index

$$\text{idx}(a,b) = (a-1) \cdot 256 + (b-1),$$

excluding index 65 535 for the identity pair. The pair space includes all 65 535 non-identity pairs.

Advanced Modes: Deep Ultra and Ultra++

- Deep Ultra: In addition to exhaustive single pairs, the compressor performs a time-limited random search over sequences of 2 to 3 transform pairs (each pair being one of the 65 535). The total number of possible sequences of up to three pairs is $65\,535 + 65\,535^2 + 65\,535^3 \approx 2.81 \times 10^{14}$.
- Ultra++: Exhaustively evaluates all 65 536 pairs (including the identity pair) and then runs the random multi-pair search for maximum coverage.

The inverse of a composed transform $T_{a_1} \circ T_{a_2} \circ \dots \circ T_{a_k}$ is $T_{a_k}^{-1} \circ \dots \circ T_{a_2}^{-1} \circ T_{a_1}^{-1}$, applied in reverse order.

Quantum-inspired additions: When Qiskit is enabled, up to 8 extra permutation transforms (indices 257–264) are generated from quantum circuits and added to the search space. They are also included in pair compositions and multi-pair searches.

4. Header Encoding

The chosen transform τ is encoded in a compact variable-length header as follows:

- Single transform T_t with $1 \leq t \leq 252$: Header = $[t-1]$ (1 byte).
- Single transform T_t with $253 \leq t \leq 256$: Header = $[254, t-253]$ (2 bytes).

- Raw (no transform): Header = [252] (1 byte).
- Pair transform $T_a \circ T_b$ with $(a,b) \in (256,256)$: Header = [253, $\lfloor \text{idx}/256 \rfloor$, $\text{idx} \bmod 256$] where $\text{idx} = (a-1) \cdot 256 + (b-1)$ (3 bytes).
- Multi-pair sequence (Deep Ultra / Ultra++): Header = [251, k , idx_1 , idx_2 , ..., idx_k] where k is the number of pairs (2 or 3) and each idx_i is a 16-bit pair index, stored big-endian. Total header length = $2 + 2k$ bytes.

Parsing is unambiguous: the first byte h_0 determines the case. No bit-stuffing or escape sequences are required.

5. Compression Backend – Dual Mode with Auto-Correction

Let $\mathcal{C} = \{C_{\text{Zstd}}, C_{\text{PAQ}}, C_{\text{raw}}\}$ be the set of backend codecs, where $C_{\text{raw}}(\mathbf{Y}) = \text{'N'} \mathbin{\text{Vert}} \mathbf{Y}$. The compressor outputs

$$C^*(\mathbf{Y}) = \arg\min_{C \in \mathcal{C}} |C(\mathbf{Y})|.$$

During decompression, the input is tried with each C^{-1} in sequence; the first successful decoding is used. If ambiguity is detected (round-trip mismatch), the system falls back to a safe mode with a one-byte marker.

Additionally, the compressor can replace the generic backend with a LZ77+Huffman pipeline. In that mode, the transformed data are first tokenised by a LZ77 parser (window 2048, min match 3), then the literals, distances, and lengths are Huffman-coded with canonical codes. An internal marker byte (0xFF for LZH, 0xFE for RLE-pre-LZH) signals this pipeline. The inverse decodes the Huffman stream and reconstructs the transformed data.

The full compression of $\mathbf{X}$ with path τ is

$\mathbf{Z} = H(\tau) \backslash; \backslash; \left(\text{marker} \backslash; \backslash; C^*(\tau(\mathbf{X})) \right) \backslash; \backslash;$
 $\text{or } H(\tau) \backslash; \backslash; \text{LZH pipeline}(\tau(\mathbf{X})).$

Decompression parses $H(\tau)$, detects the marker, decodes the payload, and returns $\tau^{-1}(\mathbf{Y})$.

6. Hybrid Dictionary Compression

Textual data can be tokenised using pre-built or dynamic dictionaries.

- Static Word Dictionary: Loaded from a merged file of 12 external dictionaries. A tokenisation function ϕ replaces known words with indices. The mapping is invertible.
- Dynamic Dictionary: The compressor partitions $\mathbf{X}$ into syntactic chunks (paragraphs, lines, sentences, words). A frequency-based dictionary is built on-the-fly, and the most frequent chunks are replaced with short codes (2 or 3 bytes).
- Line Dictionary: A greedy longest-match parser using a fixed phrase list replaces continuous text spans with phrase indices.

All dictionary modes are encapsulated with distinct magic bytes, enabling automatic detection during decompression.

7. Quantum-Inspired Transforms

When Qiskit is available, the system constructs a quantum circuit $\mathcal{Q}$ with random rotation gates. Its QASM representation is hashed to seed a permutation σ on a set of size 2^q (default $q=8$, configurable up to 49). A byte-level substitution transform is defined as

$$T_{\sigma}(x_i) = \sigma(x_i) \quad \text{if } x_i < \text{size}, \quad T_{\sigma}^{-1}(y_i) = \sigma^{-1}(y_i).$$

For $q>8$, the permutation acts on a block-by-block basis (size clamped to 1024 for performance). These transforms are unconditionally bijective and are assigned indices 257, 258, They seamlessly integrate into the pair and multi-pair search.

8. Exhaustive Self-Test and Verification

For every $t \in \{1, \dots, 256\}$ and every byte $b \in \mathbb{B}$, we verify

$$T_t^{-1}(T_t([b])) = [b].$$

For all 65 535 pairs $(a,b) \in \{1, \dots, 256\}^2 \setminus \{(256,256)\}$, we verify

$$T_a^{-1}(T_b^{-1}(T_b(T_a([b]))) = [b] \quad \text{for all } b \in \mathbb{B}.$$

Additionally, random 1000-byte blocks, empty input, and dictionary round-trips are tested. The tests pass on all paths, confirming that the compressor is 100 % lossless.

9. Conclusion

PJP-14 formalises compression as a search over a space of bijective transforms, with $1 + 256 + 65\,535 = 65\,792$ base paths and advanced modes exploring over 281 trillion combinations of up to three transform pairs. Each elementary transform is explicitly invertible, and the composition of any sequence is guaranteed invertible. The architecture guarantees bit-perfect reconstruction by design, backed by exhaustive self-verification. The inclusion of hybrid dictionaries, quantum-inspired permutations, new specialised transforms (4-byte pattern XOR, Calculus2⁶⁴), and multiple backend codecs extends the framework's applicability while preserving mathematical rigour. By treating compression as a data-driven pipeline with rigorous validation, PJP-14 achieves competitive ratios without ever sacrificing losslessness.

References

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